The goal was to analyze model interpretability and fairness using the Titanic dataset.

```
!pip install shap
```

```
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-da
```

```
import pandas as pd

url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"

df = pd.read_csv(url)

df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	E
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	

Next steps: Generate code with df New interactive sheet

```
data = df[['Pclass','Sex','Age','Fare','Survived']]

data = data.dropna()

data['Sex'] = data['Sex'].map({
```

```
    'male':0,
    'female':1
})

data.head()
```

	Pclass	Sex	Age	Fare	Survived
0	3	0	22.0	7.2500	0
1	1	1	38.0	71.2833	1
2	3	1	26.0	7.9250	1
3	1	1	35.0	53.1000	1
4	3	0	35.0	8.0500	0

Next steps: Generate code with data New interactive sheet

```
from sklearn.model_selection import train_test_split

X = data.drop('Survived', axis=1)
y = data['Survived']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier()

model.fit(X_train, y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier()
```

```
from sklearn.metrics import accuracy_score

pred = model.predict(X_test)

accuracy = accuracy_score(y_test, pred)

print("Accuracy:", accuracy)
```

```
Accuracy: 0.7692307692307693
```

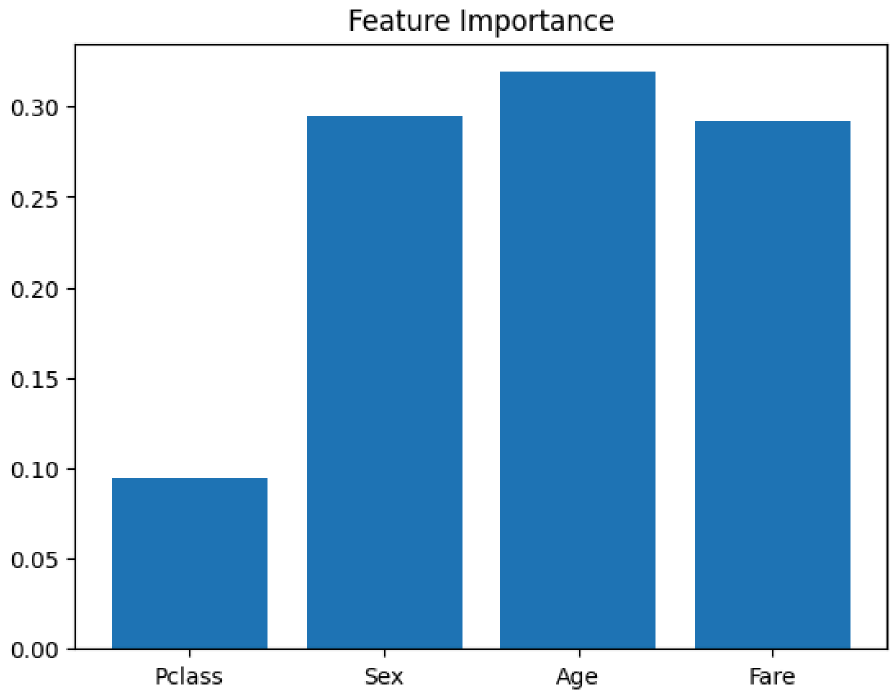
```
import matplotlib.pyplot as plt

importance = model.feature_importances_

features = X.columns

plt.bar(features, importance)

plt.title("Feature Importance")
plt.show()
```



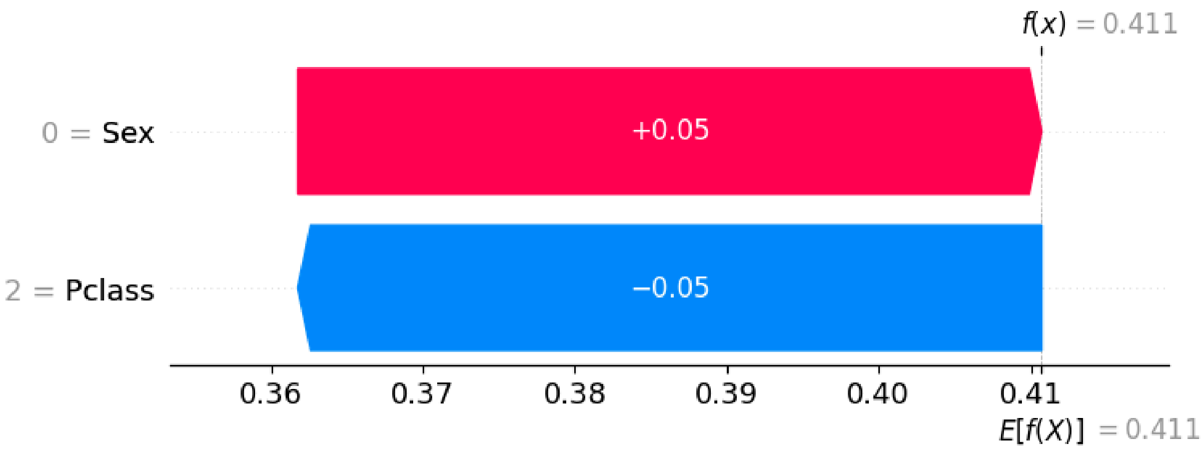

```
import shap

explainer = shap.TreeExplainer(model)

shap_values = explainer.shap_values(X_test)

shap.summary_plot(
    shap_values,
    X_test
)
```


```
shap.plots.waterfall(
    shap.Explanation(
        values=shap_values[1][0],  # SHAP values for the first instance, class 1
        base_values=explainer.expected_value[1],
        data=X_test.iloc[0].values,
        feature_names=X_test.columns.tolist()
    )
)
```



Double-click (or enter) to edit

```
male = X_test['Sex'] == 0
female = X_test['Sex'] == 1

from sklearn.metrics import accuracy_score

male_acc = accuracy_score(
    y_test[male],
    pred[male]
)

female_acc = accuracy_score(
    y_test[female],
```